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## PROFESSIONAL SUMMARY

Senior GCP Data Engineer with 4 years of experience designing and delivering scalable data pipelines and cloud-based solutions for enterprise clients including Verizon. Expertise in BigQuery, Dataflow, Apache Airflow, Apache Beam, Cloud Spanner, and Cloud Composer. Skilled in Python, SQL, PySpark, and ETL/ELT processes with a strong command of data modelling, governance, and pipeline observability. Hands-on experience applying GenAI technologies — including LangChain, LangGraph, GPT-4o, Chroma DB, and RAG — in production data projects to build intelligent query layers, data lineage assistants, and AI-powered pipeline monitoring.

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## PROFESSIONAL EXPERIENCE

### Senior Data Engineer

Prodapt Solutions Pvt Ltd

Client: Verizon

#### Project: One Network Data

Jan 2026 – Present

- Engineered and managed Airflow DAGs in Cloud Composer to orchestrate Dataflow batch and streaming pipelines, reducing manual intervention across multiple GCP data domains.
- Implemented Apache Beam transforms to load and insert large-scale data into Cloud Spanner with high throughput, ensuring transactional consistency across distributed writes.
- Built a natural language query layer over BigQuery using Scala, Play Framework, LangChain, Akka and GPT-4o, enabling analysts to retrieve pipeline insights without writing SQL.
- Developed a RAG-based data lineage assistant using Scala, Akka, Chroma DB and prompt engineering to automate data quality reports and surface anomalies from Dataflow pipeline logs.
- Integrated Cloud Storage as a staging layer for intermediate transformations before downstream delivery to BigQuery and Cloud Spanner.
- Collaborated with cross-functional teams to define data contracts and validate quality SLAs across 5+ GCP pipeline stages, ensuring schema consistency and reliability.

#### Project: Message Code Simplification

July 2025 – Dec 2025

- Architected an end-to-end Consumer Data Platform (CDP) on GCP, unifying customer data from EDW and CJCM golden sources into a single Customer Universe model.
- Built real-time ingestion pipelines using Apache Kafka, Scala, Akka and GCP Dataflow to process streams from CJCM, Cassandra, and POS systems at scale.
- Designed multi-layer curation logic enhanced with GPT-4o-assisted validation to cleanse, classify, and route raw customer data, improving downstream data quality and consistency.
- Created Scala-based BigQuery views and aggregations with GPT-4o-powered auto-summarization, delivering instant customer segment insights to Looker dashboards and ML models.
- Enforced data governance by embedding validation rules at each pipeline stage, reducing data incidents, and ensuring auditability.
- Activated unified customer profiles to Adobe Campaigns and Adobe Experience Platform, enabling personalized marketing at scale.
- Optimised end-to-end pipeline processing time by 35% through parallel execution strategies in Dataflow and efficient partitioning in BigQuery

## GenAI Training

May 2026 – Jun 2026

### Capstone Project: SGEIA (Smart Grid Energy Intelligence Assistant)

- Completed Proadapt's GenAI program covering Prompt Engineering, RAG, LangChain, LangGraph, multi-agent systems, Guardrails, DeepEval/Ragas evaluation, LangSmith/Langfuse observability, and FastAPI development.
- Built SGEIA using LangGraph StateGraph with GPT-4o, ChromaDB (hybrid BM25 + semantic RAG), XGBoost for grid stability prediction, and Isolation Forest for smart meter anomaly detection.
- Delivered a production-ready SGEIA system with FastAPI streaming backend and Streamlit dashboard, achieving >0.85 faithfulness score via DeepEval quality gates and full LangSmith observability across 5 agent nodes.

## GCP Data Engineer

Proxima Systems Pvt Ltd

Client: SnippetSentry

### Project: SnippetSentry

May 2022 – Dec 2024

- Designed and delivered end-to-end data pipelines on GCP using Apache Airflow, BigQuery, and Cloud Storage, supporting high-frequency ingestion and analytics reporting.
- Managed Cloud Storage buckets, IAM roles, and access controls to maintain data security, compliance, and auditability across environments.
- Built and optimised 15+ BigQuery datasets, tables, and SQL queries, cutting average query execution time by 40% and improving analytics turnaround.
- Proactively monitored pipeline health, identified bottlenecks, and resolved failures — significantly reducing latency and improving data reliability.
- Authored runbooks, data flow diagrams, and test coverage for 10+ deployed pipelines, enabling faster onboarding and quicker incident resolution.
- Mentored new team members on Apache Airflow best practices, standardising development workflows and accelerating team productivity.

### Project: AI Flow Mapper

Dec 2023 – Dec 2024

- Developed AI Flow Mapper, reducing manual diagram creation time by ~60% by automatically transforming requirement keywords and JSON inputs into structured mind maps and flow diagrams.
- Designed a Scala-based data ingestion layer using Play framework to parse and normalize JSON inputs, enabling dynamic generation of structured diagram outputs.
- Built backend GCP pipelines for transformation logic, storing intermediate results in Cloud Storage and serving outputs via BigQuery.
- Reduced requirements-to-diagram turnaround by 3x, enabling product and engineering teams to visualise complex workflows in minutes rather than hours.

### Project: Bentley Systems

Feb 2022 – April 2022

Client: Bentley | Role: SQL Developer

- Wrote and optimized SQL queries to extract, manipulate, and analyze data from relational databases.
- Developed complex joins, subqueries, and aggregate functions to fulfil business reporting needs.
- Designed database schemas including tables, primary and foreign keys, and indexes.
- Performed data cleaning and validation on 5+ datasets, resolving inconsistencies, and improving data accuracy for downstream reporting.
- Supported reporting dashboards by writing backend SQL queries for data sourcing and summaries.

## TECHNICAL SKILLS

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|----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Cloud Platforms</b>     | Google Cloud Platform (BigQuery, Cloud Storage, Dataflow, DataProc, Pub/Sub, Cloud Spanner, Cloud Composer), Snowflake |
| <b>Data Tools</b>          | Apache Airflow, Apache Beam, Apache Spark, Databricks, MongoDB, Databuck,,Hadoop (HDFS, Hive, YARN)                    |
| <b>AI / GenAI</b>          | LangGraph, LangChain, OpenAI GPT-4o, Chroma DB, RAG, Prompt Engineering, LangSmith, DeepEval                           |
| <b>ML &amp; Evaluation</b> | XGBoost, Isolation Forest, SHAP                                                                                        |
| <b>BI &amp; Automation</b> | Power BI, Google Cloud Console, Looker                                                                                 |
| <b>Programming</b>         | Python, SQL, JavaScript, PySpark, Scala, Basic Java                                                                    |
| <b>Frameworks</b>          | LangChain, LangGraph, FastAPI, Play Framework, Akka                                                                    |
| <b>Others</b>              | Agile, GIT Labs, JIRA                                                                                                  |

## CERTIFICATIONS

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- Python (Core and Advanced) — National Skill Development Corporation
- Introduction to Artificial Intelligence — LinkedIn
- GenAI Program — Prodapt Solutions

## EDUCATION

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### **Bachelor of Technology – Electronics & Communication Engineering**

Andhra University, Visakhapatnam

2018 – 2022

### **Intermediate (MPC)**

Sri Chaitanya Junior College, Visakhapatnam

2016 – 2018